

## Section 1: Upload Files from Multiple Folders

This section uploads files from three different folders.

```
from google.colab import files

print("Please upload files from 'Mental Health VRChat Stories' folder")
uploaded_mental_health = files.upload()

print("Please upload files from 'Military VRChat Stories' folder")
uploaded_military = files.upload()

print("Please upload files from 'Queer VRChat Stories' folder")
uploaded_queer = files.upload()
```


Please upload files from 'Mental Health VRChat Stories' folder

No file chosen

**Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.**

Saving 1. Girl in VRChat Has Synesthesia.txt to 1. Girl in VRChat Has Synesthesia.txt  
Saving 2. Person in VR Has Dissociative Identity Disorder.txt to 2. Person in VR Has Dissociative Identity Disorder.txt  
Saving 3. Living With an Extremely Rare Eating Disorder.txt to 3. Living With an Extremely Rare Eating Disorder.txt  
Saving 4. Young Girl Talks About Being Bullied for Having Autism.txt to 4. Young Girl Talks About Being Bullied for Having Autism.txt  
Saving 5. Girl in VRChat Talks About Her Tourette\_s.txt to 5. Girl in VRChat Talks About Her Tourette\_s.txt  
Saving 6. Guy Talks About Surviving a Corrupt Mental Hospital.txt to 6. Guy Talks About Surviving a Corrupt Mental Hospital.txt  
Saving 7. Meeting a Diagnosed Sociopath in VRChat.txt to 7. Meeting a Diagnosed Sociopath in VRChat.txt  
Saving 8. What It\_s Like to Know When You\_re Going to Die.txt to 8. What It\_s Like to Know When You\_re Going to Die.txt  
Saving 9. He Was Trapped in Isolation for 153 Days as a Child.txt to 9. He Was Trapped in Isolation for 153 Days as a Child.txt  
Saving 10. The Most Terrifying Mental Disorder (VR Interview).txt to 10. The Most Terrifying Mental Disorder (VR Interview).txt  
Saving 11. I Might Not See Her Alive Again.txt to 11. I Might Not See Her Alive Again.txt  
Saving 12. We Need to Talk About This More.txt to 12. We Need to Talk About This More.txt  
Saving 13. I Met an Ex-Incel - What He Had to Say was Incredible.txt to 13. I Met an Ex-Incel - What He Had to Say was Incredible.txt  
Saving 14. VRChat Stories that Make Me VERY Concerned....txt to 14. VRChat Stories that Make Me VERY Concerned....txt  
Saving 15. Surviving Drug Addiction at Age 15.txt to 15. Surviving Drug Addiction at Age 15.txt  
Saving 16. Real Life Hypnotist Teaches You to Defend Your Brain.txt to 16. Real Life Hypnotist Teaches You to Defend Your Brain.txt  
Saving 16. Surviving Drug Addiction at Age 15.txt to 16. Surviving Drug Addiction at Age 15.txt  
Saving 17. They Worked For a Corrupt Disability Care Facility - Here\_s What They Saw.txt to 17. They Worked For a Corrupt Disability  
Saving 18. Her Last Recording.txt to 18. Her Last Recording.txt  
Please upload files from 'Military VRChat Stories' folder

No file chosen

**Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.**

Saving 1. 1-05-57 NOW PLAYING Volunteer Militant Describes Craziest Battles Against ISIS.txt to 1. 1-05-57 NOW PLAYING Volunteer Militant Describes Craziest Battles Against ISIS.txt  
Saving 2. This Marine Fought Russians in Ukraine - Here\_s What He Saw.txt to 2. This Marine Fought Russians in Ukraine - Here\_s What He Saw.txt  
Saving 3. Veteran Talks About the Dark Side of the Military.txt to 3. Veteran Talks About the Dark Side of the Military.txt  
Saving 4. Airstrike Geospatial Intelligence Operator Tells His Story in VR.txt to 4. Airstrike Geospatial Intelligence Operator Tells His Story in VR.txt  
Saving 5. US Military Cyber Warfare is Absolutely Unhinged (VR Interview).txt to 5. US Military Cyber Warfare is Absolutely Unhinged (VR Interview).txt  
Saving 6. Some Sussy Submarine Stories.txt to 6. Some Sussy Submarine Stories.txt  
Saving 7. Guy Talks About Surviving a Corrupt Military Base.txt to 7. Guy Talks About Surviving a Corrupt Military Base.txt  
Saving 8. This Air Force Veteran Uses VR to Treat Her PTSD.txt to 8. This Air Force Veteran Uses VR to Treat Her PTSD.txt  
Saving 9. Guy in VR Describes Helping Evacuate Afghanistan.txt to 9. Guy in VR Describes Helping Evacuate Afghanistan.txt  
Saving 10. Meeting South Korean Soldier in VR.txt to 10. Meeting South Korean Soldier in VR.txt  
Saving 11. He Fought Real-Life Pirates With the German Special Forces.txt to 11. He Fought Real-Life Pirates With the German Special Forces.txt  
Saving 12. How the Taliban Weaponized Chocolate (& Other Stories).txt to 12. How the Taliban Weaponized Chocolate (& Other Stories).txt  
Saving 13th videox.txt to 13th videox.txt  
Saving 14. The Coast Guard is Absolutely Unhinged.txt to 14. The Coast Guard is Absolutely Unhinged.txt  
Please upload files from 'Queer VRChat Stories' folder

No file chosen

**Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.**

Saving 1. Welcome to the USA, Where Torturing Children is Legal.txt to 1. Welcome to the USA, Where Torturing Children is Legal.txt  
Saving 2. Based Sexuality.txt to 2. Based Sexuality.txt  
Saving 3. The Most Important Video I\_ll Ever Make.txt to 3. The Most Important Video I\_ll Ever Make.txt  
Saving 4. Being Gay in a Country that Wants LGBTQ People Dead.txt to 4. Being Gay in a Country that Wants LGBTQ People Dead.txt  
Saving 5. He Came Out as Trans in High School - It Was Hell.txt to 5. He Came Out as Trans in High School - It Was Hell.txt  
Saving 6. She Was Born Different - But Nobody Told Her.txt to 6. She Was Born Different - But Nobody Told Her.txt  
Saving 7. Girl Talks About Being Gay in Catholic School.txt to 7. Girl Talks About Being Gay in Catholic School.txt  
Saving 8. Doctors Denied Her Condition - So She Proved Them Wrong.txt to 8. Doctors Denied Her Condition - So She Proved Them Wrong.txt  
Saving 9. He Is a Trans Athlete in High School - It\_s Rough.txt to 9. He Is a Trans Athlete in High School - It\_s Rough.txt  
Saving 10. From Faith to Freedom - A Story of Religious Trauma.txt to 10. From Faith to Freedom - A Story of Religious Trauma.txt  
Saving 11. Saving Trans Lives in VR - The VRChat Trans Academy.txt to 11. Saving Trans Lives in VR - The VRChat Trans Academy.txt  
Saving 12. Her Father is a Neo-Nazi - But She\_s Trans.txt to 12. Her Father is a Neo-Nazi - But She\_s Trans.txt

## Section 2: Install Required Packages

This section ensures all necessary packages are installed.

```
!pip install nltk gensim vaderSentiment textblob matplotlib pandas pyLDAvis bertopic
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.10/dist-packages (3.8.1)
Requirement already satisfied: gensim in /usr/local/lib/python3.10/dist-packages (4.3.2)
Requirement already satisfied: vaderSentiment in /usr/local/lib/python3.10/dist-packages (3.3.2)
Requirement already satisfied: textblob in /usr/local/lib/python3.10/dist-packages (0.17.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.10/dist-packages (3.7.1)
Requirement already satisfied: pandas in /usr/local/lib/python3.10/dist-packages (2.0.3)
Requirement already satisfied: pyLDAvis in /usr/local/lib/python3.10/dist-packages (3.4.1)
Requirement already satisfied: bertopic in /usr/local/lib/python3.10/dist-packages (0.16.2)
Requirement already satisfied: click in /usr/local/lib/python3.10/dist-packages (from nltk) (8.1.7)
Requirement already satisfied: joblib in /usr/local/lib/python3.10/dist-packages (from nltk) (1.4.2)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.10/dist-packages (from nltk) (2023.12.25)
Requirement already satisfied: tqdm in /usr/local/lib/python3.10/dist-packages (from nltk) (4.66.4)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.10/dist-packages (from gensim) (1.25.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.10/dist-packages (from gensim) (1.11.4)
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.10/dist-packages (from gensim) (6.4.0)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (1.2.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (1.4.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (24.0)
Requirement already satisfied: pillow>=6.2.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas) (2023.4)
Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.10/dist-packages (from pandas) (2024.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.10/dist-packages (from pyLDAvis) (3.1.4)
Requirement already satisfied: NumExpr in /usr/local/lib/python3.10/dist-packages (from pyLDAvis) (2.10.0)
Requirement already satisfied: funcy in /usr/local/lib/python3.10/dist-packages (from pyLDAvis) (2.0)
Requirement already satisfied: scikit-learn>=1.0.0 in /usr/local/lib/python3.10/dist-packages (from pyLDAvis) (1.2.2)
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages (from pyLDAvis) (67.7.2)
Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.10/dist-packages (from bertopic) (0.8.33)
Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.10/dist-packages (from bertopic) (0.5.6)
Requirement already satisfied: sentence-transformers>=0.4.1 in /usr/local/lib/python3.10/dist-packages (from bertopic) (2.7.0)
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.10/dist-packages (from bertopic) (5.15.0)
Requirement already satisfied: cython<3,>=0.27 in /usr/local/lib/python3.10/dist-packages (from hdbscan>=0.8.29->bertopic) (0.29.3)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.10/dist-packages (from plotly>=4.7.0->bertopic) (8.3.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in /usr/local/lib/python3.10/dist-packages (from scikit-learn>=1.0.0->pyLDAvis) (3.2.0)
Requirement already satisfied: transformers<5.0.0,>=4.34.0 in /usr/local/lib/python3.10/dist-packages (from sentence-transformers>=0.4.1->bertopic) (4.41.0)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.10/dist-packages (from sentence-transformers>=0.4.1->bertopic) (2.4.0)
Requirement already satisfied: huggingface-hub>=0.15.1 in /usr/local/lib/python3.10/dist-packages (from sentence-transformers>=0.4.1->bertopic) (0.24.0)
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.10/dist-packages (from umap-learn>=0.5.0->bertopic) (0.58.1)
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.10/dist-packages (from umap-learn>=0.5.0->bertopic) (0.5.1)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (from Jinja2->pyLDAvis) (2.1.5)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests->vaderSentiment) (3.3.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests->vaderSentiment) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests->vaderSentiment) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests->vaderSentiment) (2024.7.4)
Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from huggingface-hub>=0.15.1->sentence-transformers) (3.13.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.10/dist-packages (from huggingface-hub>=0.15.1->sentence-transformers) (2024.9.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.10/dist-packages (from huggingface-hub>=0.15.1->sentence-transformers) (6.0.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.10/dist-packages (from huggingface-hub>=0.15.1->sentence-transformers) (4.11.0)
Requirement already satisfied: llvmlite<0.42,>=0.41.0dev0 in /usr/local/lib/python3.10/dist-packages (from numba>=0.51.2->umap-learn) (0.42.0)
Requirement already satisfied: sympy in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (1.12.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (3.2.1)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.1.105 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (12.1.105)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.1.105 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (12.1.105)
Requirement already satisfied: nvidia-cublas-cu12==12.1.105 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (12.1.105)
Requirement already satisfied: nvidia-cudnn-cu12==8.7.0 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (8.7.0)
Requirement already satisfied: nvidia-nccl-cu12==2.19.3 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (2.19.3)
Requirement already satisfied: nvidia-nvtx-cu12==12.1.105 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (12.1.105)
Requirement already satisfied: triton==2.1.0 in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (2.1.0)
Requirement already satisfied: sympy in /usr/local/lib/python3.10/dist-packages (from torch>=1.11.0->sentence-transformers>=0.4.1->bertopic) (1.12.0)
Requirement already satisfied: mpmath in /usr/local/lib/python3.10/dist-packages (from sympy) (1.3.0)
Requirement already satisfied: MarkupSafe in /usr/local/lib/python3.10/dist-packages (from Jinja2->pyLDAvis) (2.1.5)
Requirement already satisfied: Pygments in /usr/local/lib/python3.10/dist-packages (from pygments) (2.18.0)
Requirement already satisfied: MarkupSafe in /usr/local/lib/python3.10/dist-packages (from Jinja2->pyLDAvis) (2.1.5)
```

## Section 3: Preprocess the Text

This section preprocesses the text, removing stop words, tokenizing, lemmatizing, and handling collocations.

```

import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
import re
from gensim.models import Phrases
from gensim.models.phrases import Phraser

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

def preprocess_text(text):
    text = re.sub(r'^\w\s', '', text.lower())
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stopwords.words('english')]
    lemmatizer = WordNetLemmatizer()
    tokens = [lemmatizer.lemmatize(token) for token in tokens]
    return tokens

def preprocess_uploaded_files(uploaded_files):
    preprocessed_texts = []
    for file_name, content in uploaded_files.items():
        preprocessed_texts.append(preprocess_text(content.decode('utf-8')))
    return preprocessed_texts

preprocessed_texts_mental_health = preprocess_uploaded_files(uploaded_mental_health)
preprocessed_texts_military = preprocess_uploaded_files(uploaded_military)
preprocessed_texts_queer = preprocess_uploaded_files(uploaded_queer)

combined_preprocessed_texts = preprocessed_texts_mental_health + preprocessed_texts_military + preprocessed_texts_queer

phrases = Phrases(combined_preprocessed_texts, min_count=5, threshold=10)
bigram = Phraser(phrases)

combined_preprocessed_texts = [bigram[text] for text in combined_preprocessed_texts]

def save_preprocessed_data(preprocessed_texts, output_file):
    with open(output_file, 'w') as file:
        for text in preprocessed_texts:
            file.write(' '.join(text) + "\n")

output_file = "combined_preprocessed_transcripts.txt"
save_preprocessed_data(combined_preprocessed_texts, output_file)

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...

```

Double-click (or enter) to edit

## ✓ Section 4: Perform Topic Modeling with Gensim LDA

This section uses Gensim's LDA model to perform topic modeling and visualizes the topics using pyLDavis.

```

import gensim
from gensim import corpora
from gensim.models import LdaModel

dictionary = corpora.Dictionary(combined_preprocessed_texts)

```

```
corpus = [dictionary.doc2bow(text) for text in combined_preprocessed_texts]
```

```
def compute_coherence_values(dictionary, corpus, texts, limit, start=5, step=2):
    coherence_values = []
    model_list = []
    for num_topics in range(start, limit, step):
        model = LdaModel(corpus=corpus, num_topics=num_topics, id2word=dictionary, passes=15, random_state=42)
        model_list.append(model)
        coherencemodel = gensim.models.CoherenceModel(model=model, texts=texts, dictionary=dictionary, coherence='c_v')
        coherence_values.append(coherencemodel.get_coherence())
    return model_list, coherence_values

start = 5
limit = 20
step = 2

model_list, coherence_values = compute_coherence_values(dictionary, corpus, combined_preprocessed_texts, limit, start, step)

print(f'Coherence Score: {max(coherence_values)}')
```

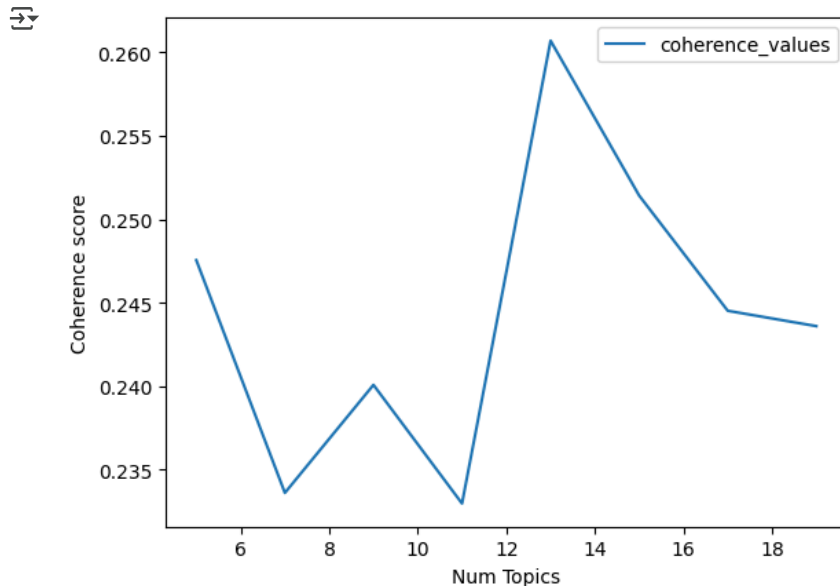
↗ Coherence Score: 0.26070014583183687

```
import matplotlib.pyplot as plt

x = range(start, limit, step)
plt.plot(x, coherence_values)
plt.xlabel("Num Topics")
plt.ylabel("Coherence score")
plt.legend(["coherence_values"], loc='best')
plt.show()

optimal_model = model_list[coherence_values.index(max(coherence_values))]
optimal_num_topics = start + coherence_values.index(max(coherence_values)) * step

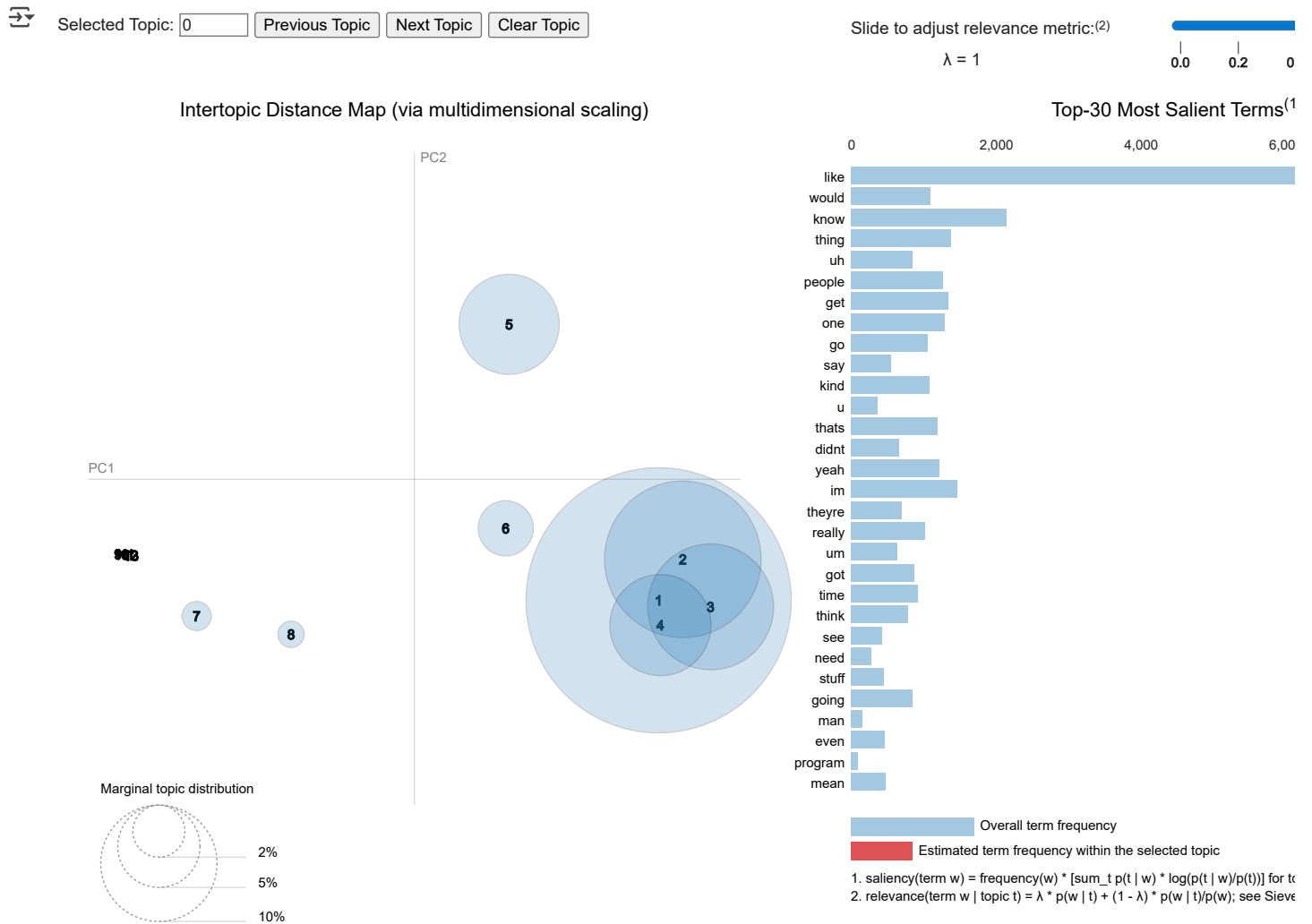
print(f'Optimal number of topics: {optimal_num_topics}')
```



Optimal number of topics: 13

```
import pyLDAvis.gensim_models as gensimvis
import pyLDAvis

vis_data = gensimvis.prepare(optimal_model, corpus, dictionary)
pyLDAvis.display(vis_data)
```



## Section 5: Perform Topic Modeling with BERTopic

This section uses BERTopic to perform topic modeling.

```
from bertopic import BERTopic
combined_texts = [' '.join(text) for text in combined_preprocessed_texts]
topic_model = BERTopic()
topics, probs = topic_model.fit_transform(combined_texts)
bertopic_results = topic_model.get_topic_info()

print("BERTopic Results:")
print(bertopic_results)
```

```

/usr/local/lib/python3.10/dist-packages/ipykernel/ipkernel.py:283: DeprecationWarning: `should_run_async` will not call `transform_c
and should_run_async(code)
/usr/local/lib/python3.10/dist-packages/skimage/util/dtype.py:27: DeprecationWarning: `np.bool8` is a deprecated alias for `np.bool_
np.bool8: (False, True),

modules.json: 100% 349/349 [00:00<00:00, 8.18kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 4.47kB/s]

README.md: 100% 10.7k/10.7k [00:00<00:00, 408kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 1.88kB/s]

config.json: 100% 612/612 [00:00<00:00, 43.0kB/s]

model.safetensors: 100% 90.9M/90.9M [00:00<00:00, 152MB/s]

tokenizer_config.json: 100% 350/350 [00:00<00:00, 19.9kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 4.03MB/s]

tokenizer.json: 100% 466k/466k [00:00<00:00, 15.8MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 5.96kB/s]

1_Pooling/config.json: 100% 190/190 [00:00<00:00, 12.1kB/s]

/usr/local/lib/python3.10/dist-packages/hdbscan/hdbscan_.py:1170: DeprecationWarning: `alltrue` is deprecated as of NumPy 1.25.0, ar
self._all_finite = is_finite(X)
BERTopic Results:
  Topic  Count      Name  \
0      -1      15  -1_like_im_get_people
1       0      17  0_like_know_thing_people
2       1      13  1_like_know_yeah_get

      Representation  \
0 [like, im, get, people, thing, know, would, on...
1 [like, know, thing, people, im, would, kind, r...
2 [like, know, yeah, get, one, thats, im, uh, go...

      Representative_Docs
0 [thats hell title let talk verify guy legit th...
1 [reality hit youre like away home im away frie...
2 [hello everyone azeal vr_chat today two vetera...

```

## ✓ Section 6: Compare Results and Document

This section documents and compares the results of LDA and BERTopic.

```

import pandas as pd

lda_topics = optimal_model.print_topics(num_words=10)
results = []

for idx, topic in lda_topics:
    words = topic.split(' + ')
    words = [word.split('*')[1].strip('') for word in words]
    texts = [combined_texts[i] for i in range(len(topics)) if topics[i] == idx][:11]
    results.append([f"Topic {idx}", ', '.join(words), ' | '.join(texts)])

df_results = pd.DataFrame(results, columns=["Topic label with topic prevalence", "Most relevant words", "Two of the most exemplary text"])

print("LDA Topic Modeling Results:")
print(df_results)

print("\nBERTopic Results:")
print(bertopic_results)

```

```

LDA Topic Modeling Results:
  Topic label with topic prevalence  \
0                                Topic 0
1                                Topic 1
2                                Topic 2

```

```

3 Topic 3
4 Topic 4
5 Topic 5
6 Topic 6
7 Topic 7
8 Topic 8
9 Topic 9
10 Topic 10
11 Topic 11
12 Topic 12

```

```

Most relevant words \
0 individual, would, agency, program, client, st...
1 like, know, would, really, get, im, thing, peo...
2 like, know, would, thing, uh, im, people, get,...
3 like, yeah, know, one, get, thats, would, thin...
4 like, know, thing, people, get, im, time, yeah...
5 like, thing, people, uh, get, go, theyre, pers...
6 like, know, get, im, thing, yeah, uh, thats, w...
7 like, know, people, thing, kind, would, one, r...
8 like, know, im, get, one, go, thats, thing, ki...
9 like, know, thing, im, would, one, get, kind, ...
10 like, know, people, thing, one, get, __, go, g...
11 like, uh, know, um, thing, yeah, get, people, ...
12 marine, refugee, u, would, like, know, uh, lef...

```

```

Two of the most exemplary text from VR chat (each video at least one)
0 synesthesia basically mean like crosssenses he...
1 tactiqio free youtube transcript surviving dru...
2
3
4
5
6
7
8
9
10
11
12

```

```

BERTopic Results:
Topic Count Name \
0 -1 15 -1_like_im_get_people
1 0 17 0_like_know_thing_people
2 1 13 1_like_know_yeah_get

```

```

Representation \
0 [like, im, get, people, thing, know, would, on...
1 [like, know, thing, people, im, would, kind, r...
2 [like, know, yeah, get, one, thats, im, uh, go...

```

## ✓ Section 7: Perform Sentiment Analysis and Plot Sentiments

This section performs sentiment analysis and plots the results.

```

from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
from textblob import TextBlob

```

```

def sentiment_analysis(text):
    vader_analyzer = SentimentIntensityAnalyzer()
    vader_score = vader_analyzer.polarity_scores(text)
    textblob_score = TextBlob(text).sentiment
    return vader_score, textblob_score

```

```

combined_text = ' '.join([' '.join(text) for text in combined_preprocessed_texts])

```

```

vader_score, textblob_score = sentiment_analysis(combined_text)
print(f"VADER: {vader_score}")
print(f"TextBlob: {textblob_score}")

```

```

/usr/local/lib/python3.10/dist-packages/ipykernel/ipkernel.py:283: DeprecationWarning: `should_run_async` will not call `transform_c
and should_run_async(code)
VADER: {'neg': 0.115, 'neu': 0.627, 'pos': 0.259, 'compound': 1.0}

```

TextBlob: Sentiment(polarity=0.08791589012547747, subjectivity=0.500563730427613)

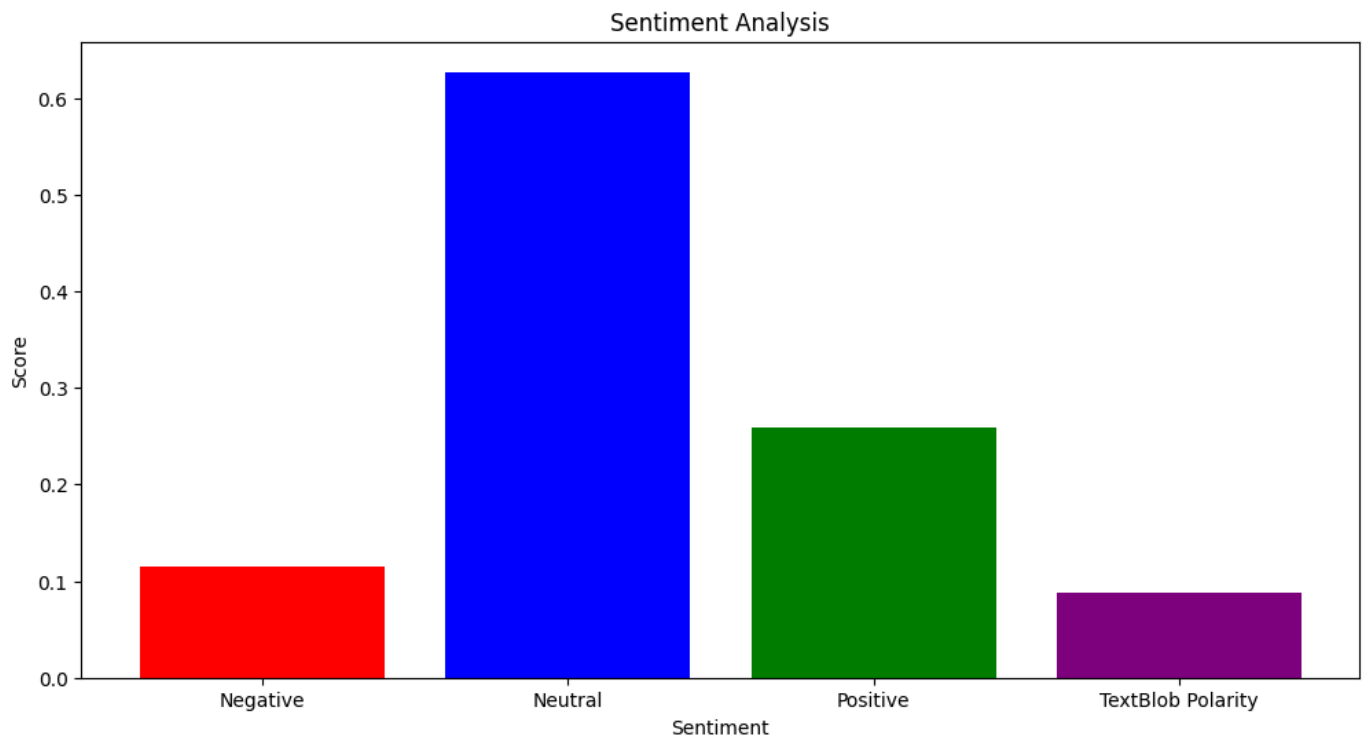
```
import matplotlib.pyplot as plt
```

```
def plot_sentiments(vader_score, textblob_score):
    labels = ['Negative', 'Neutral', 'Positive', 'TextBlob Polarity']
    vader_values = [vader_score['neg'], vader_score['neu'], vader_score['pos'], textblob_score.polarity]
```

```
    plt.figure(figsize=(12, 6))
    plt.bar(labels, vader_values, color=['red', 'blue', 'green', 'purple'])
    plt.title('Sentiment Analysis')
    plt.xlabel('Sentiment')
    plt.ylabel('Score')
    plt.show()
```

```
plot_sentiments(vader_score, textblob_score)
```

 /usr/local/lib/python3.10/dist-packages/ipykernel/ipkernel.py:283: DeprecationWarning: `should\_run\_async` will not call `transform\_c` and `should\_run\_async` (code)



## ✓ Section 8: Perform Emotion Analysis and Plot Emotions

This section uses the NRC Emotion Lexicon file to perform emotion analysis and plot the results.

```
import pandas as pd
```

```
uploaded_nrc = files.upload()
nrc_file_name = list(uploaded_nrc.keys())[0]
```

```
nrc = pd.read_csv(nrc_file_name,
                  names=["word", "emotion", "association"],
                  sep='\t')
```

```
nrc = nrc[nrc['association'] == 1]
```

```
def emotion_analysis(text):
    tokens = text.split()
    emotions = nrc[nrc['word'].isin(tokens)]
    emotion_counts = emotions['emotion'].value_counts().reset_index()
    emotion_counts.columns = ['emotion', 'count']
```

```

return emotion_counts

emotion_counts = emotion_analysis(combined_text)

import matplotlib.pyplot as plt

def plot_emotions(emotion_counts):
    plt.figure(figsize=(12, 6))
    plt.bar(emotion_counts['emotion'], emotion_counts['count'], color='skyblue')
    plt.title('Emotion Analysis')
    plt.xlabel('Emotion')
    plt.ylabel('Count')
    plt.xticks(rotation=45)
    plt.show()

plot_emotions(emotion_counts)

```

→ /usr/local/lib/python3.10/dist-packages/ipykernel/ipkernel.py:283: DeprecationWarning: `should\_run\_async` will not call `transform\_c` and should\_run\_async(code)

Choose Files No file chosen

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Saving NRC-Emotion-Lexicon-Wordlevel-v0.92.txt to NRC-Emotion-Lexicon-Word level-v0.92 (1).txt

